

Multi-dimensional Verification Methodology of Ionospheric Gradient Observation during Plasma Bubble Events in the Brazilian Region

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ABSTRACT

Ionospheric activity in equatorial regions is known to be significantly more variable and intense than in mid-latitude regions. Prior to the initiation of Ground-Based Augmentation System (GBAS) service in the Brazilian region, it is necessary to make provisions for being sufficiently robust to all possible ionospheric anomalies through development of an ionospheric anomaly threat model. In the Brazilian region, ionospheric spatial gradients larger than the upper bounds of the Conterminous U.S (CONUS) threat model developed for the mid-latitude region are frequently observed in the presence of Equatorial Plasma Bubbles (EPBs). The higher bounds in the resulting ionospheric threat model have a significant effect on performance, availability, and the eventual approval of the system. Thus, verification of observed ionospheric spatial gradients is important since observations verified to be due to EPBs will be used to determine the upper bound of the ionospheric anomaly threat model for GBAS in Brazil.

This paper applies a two-phase procedure of validating extremely large ionospheric gradients caused by low-latitude ionospheric anomaly events in the Brazilian region. This multi-dimensional approach rules out the possibility of receiver-instigated events, single satellite faults, and post-processing errors creating an apparent gradient that is nonexistent. However, the relatively smaller scale of EPBs and the sparse distribution of the Brazilian network stations makes it difficult to use the existing validation methods developed for mid-latitudes. Thus, this paper also proposes a new verification methodology based on the time-step method in order to augment the two-phase validation procedure. This methodology utilizes multiple satellites and stations that exhibit a similar trend of ionospheric delay to that of the threat candidate. The time-step method enables us to estimate ionospheric gradients over any short baseline distance and thus compensates for the lack of ionospheric observability caused by a sparse distribution of GPS reference stations. It visualizes the equatorial anomaly event in both the time domain and spatial domain by

utilizing all available sources including regional ionospheric delay maps, IPP tracks, the motion of EPBs, the location of stations, and the occurrence time of large gradients.

Using this procedure, we verified an extreme ionospheric gradient of 501.2 mm/km observed at reference stations SAVO and SSA1 viewing PRN 21 on 31 December 2013. Simultaneous ionospheric observation data from another satellite passing across the same EPB region was as high as about 360 mm/km. Six station-satellite pairs exhibited similar trend of ionospheric delays and observed severe ionospheric gradients above 300 mm/km using the time-step based techniques. Threat event visualization supported the fact that satellite-station pairs spread over several hundreds of kilometers in east-west direction were impacted by the same EPB moving eastward. These factors give us complete support to confirm that the extreme ionospheric gradient observed here is caused by a real ionospheric anomaly and not due to measurement errors or other faults. In addition to the particular case presented in this paper, we have also completed verification of other extreme EPB events whose gradients were larger than 500 mm/km. However, the full description of other events is beyond the scope of the paper.

1.0. INTRODUCTION

Ground-Based Augmentation Systems (GBAS) that enhance the performance of the Global Positioning System (GPS) must make provisions for being sufficiently robust to ionospheric anomalies and irregularities. A Honeywell SLS-4000 GBAS ground facility installed at the Galeão International Airport in Brazil is currently configured with the Conterminous U.S (CONUS) threat model [1]. However, ionospheric activity in equatorial regions (within 25 degrees latitude of the geomagnetic equator) is known to be significantly more variable and more intense than what is encountered (and has been extensively studied) in mid-latitude regions such as CONUS [1]. Models of nominal and threatening

ionospheric behavior developed for mid-latitude regions do not apply in equatorial regions. This motivated the development of a new model for Brazil based upon analyzing several years of data collected in this region over the most recent solar maximum (from 2011 to 2014).

Preliminary results from accessing GBAS ionospheric behavior using Brazilian GPS observation data show that the occurrence of ionospheric spatial gradients above 500 mm/km occurs often enough to be of concern under the presence of “Equatorial Plasma Bubbles (EPBs)” of charged-particle depletion [2]. This implies that the upper bound on spatial gradients for the Brazilian ionospheric anomaly threat model needs to be elevated to a considerably higher value compared to that of the CONUS model. Since the new threat model bound can have a significant effect on the performance and availability of the system, an exhaustive validation of the observed ionospheric anomalies is required to confirm that these are actual ionospheric events and not faulty measurements caused by receiver glitches or post-processing errors.

The procedure of validating ionospheric spatial gradients consists of two phases. In the first phase, validation is performed on two stations viewing one satellite that observes an extreme gradient using the method known as “manual validation” of Long Term Ionospheric Anomaly Monitoring (LTIAM) software [3]. This process re-examines the dual-frequency carrier-phase estimate to confirm that they reflect actual ionospheric event. However, the EPBs, which occur often during the post sunset hours at the peaks of the seasonal cycle around equinoxes, strongly coincide with amplitude and phase scintillations. These rapidly varying irregularities cause frequent outages and cycle slips on carrier-phase measurements and consequently trigger excessive data outliers and short-arc removals during the LTIAM process [3]. Thus, dual-frequency measurement data recovery has been added to the “manual validation” process. This new step turns off the short-arc and outlier removal process in order to prevent removing large ionospheric spatial gradient data. Then, the recovered carrier-derived estimate is inspected using “manual validation with single frequency measurements” for verification.

The second phase of the validation procedure identifies faulty gradient candidates caused by receiver-instigated events or post-processing error using other stations and other satellites. The methods used in this procedure, “station-wide checks” and “satellite-wide checks,” have been used to verify severe ionospheric gradients caused by large-scale ionospheric enhancements in mid-latitude regions [1]. These are still useful, but they require modification and augmentation due to the nature of “plasma bubbles.” “Station-wide and satellite-wide checks” utilize nearby station pairs and other satellites which pass through the same region as the observed ionospheric event to confirm that a single receiver or a

single satellite fault did not generate an erroneous observation. However, the relatively smaller size of EPBs (with widths of a few to several tens of kilometers in the east-west direction) and the sparse distribution of the Brazilian network stations make it difficult to use the methods derived for mid-latitudes.

This paper presents a new methodology of ionospheric observation verification that augments the existing validation methods. A key technique in this methodology is the “time-step method” (developed in [4]), which compensates for the lack of ionospheric observability by measuring ionospheric gradients using a single satellite and a single station over time. The two-phase validation procedure with the revised multi-dimensional approach (taking into account any false anomaly case caused by all possible sources, such as receiver-instigated events, a single satellite fault, post-processing errors, and so on) enables us to scrutinize extreme ionospheric gradients more carefully in the Brazilian region. Section 2.0 introduces the data used for the Brazilian ionospheric analysis. Section 3.0 describes the results and limitations of the two-phase validation procedure with the existing validation methods. In Section 4.0, a new validation methodology based on the “time-step method” is explained in detail. Section 5.0 presents the results from a case study verified using the threat visualization procedure. Section 6.0 reports the conclusions of this paper.

2.0. DATA FROM BRAZIL

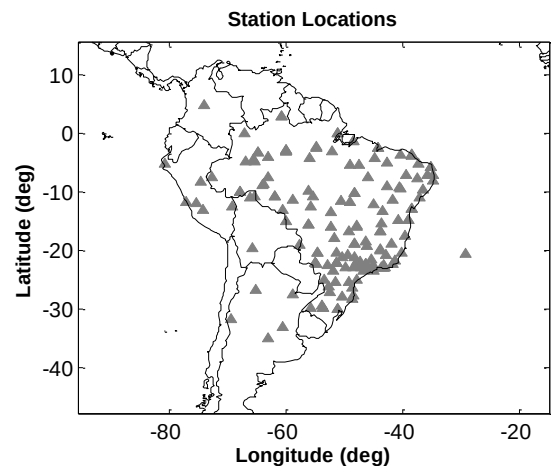


Figure 1. Location of the Brazilian GPS reference stations operated by RBMC, LISN, CIGALA, ICEA, SIPEG, and Honeywell as of 2014.

Dual-frequency code and carrier-phase GPS measurements (L1 at 1575.42MHz and L2 at 1227.60MHz) have been obtained from networks of Brazilian GPS reference stations to estimate ionospheric delay and ionospheric spatial gradient using LTIAM software [3]. These reference stations are operated by the Brazilian Network for Continuous GPS Monitoring (RBMC), the Low-latitude Ionospheric Sensor Network (LISN), the Challenging the Solar Maximum in Brazil

with PolARxS (CIGALA), the Brazilian Airspace Control Institute (ICEA), the Integrated Positioning System for Geodynamic Studies (SIPEG), and Honeywell International. These networks provide dual-frequency GPS data in Receiver Independent Exchange (RINEX) format with sampling periods from 1 to 30 sec. Figure 1 shows the locations of the Brazilian GPS reference stations, of which 185 stations are operational as of 2014.

We processed data of 32 pre-selected active ionospheric days during the peak of the current solar cycle from April 2011 to March 2014 for detailed investigation of ionospheric spatial decorrelation in the Brazilian region. This dataset consists of 22 scintillating, 2 non-scintillating, and 8 stormy days (based on geomagnetic index, K_p and Dst). Using this selected dataset, we investigated GBAS ionospheric spatial decorrelation in the Brazilian region.

Table 1. Dates analyzed and corresponding ionosphere conditions

Group Type	Date (mm/dd/yyyy)
Scintillating days	11/06/2013
Scintillating days	11/07/2013
Scintillating days	11/14/2013
Scintillating days	11/15/2013
Scintillating days	11/22/2013
Scintillating days	12/01/2013
Scintillating days	12/18/2013
Scintillating days	12/19/2013
Scintillating days	12/26/2013
Scintillating days	12/31/2013
Scintillating days	01/09/2014
Scintillating days	01/10/2014
Scintillating days	01/19/2014
Scintillating days	01/20/2014
Scintillating days	01/27/2014
Scintillating days	01/28/2014
Scintillating days	02/04/2014

Scintillating days	02/05/2014
Scintillating days	02/24/2014
Scintillating days	02/25/2014
Scintillating days	03/06/2014
Scintillating days	03/07/2014
Non-scintillating days	04/18/2012
Non-scintillating days	04/19/2012
Stormy days	09/17/2011
Stormy days	09/27/2011
Stormy days	10/01/2012
Stormy days	10/08/2012
Stormy days	06/01/2013
Stormy days	06/28/2013
Stormy days	07/14/2013
Stormy days	04/12/2014

3.0. CONCERNS WITH EXISTING VALIDATION METHODS

3.1 FIRST-PHASE VALIDATION PROCEDURE: EXTREME IONOSPHERIC GRADIENT

To investigate GBAS ionospheric behavior in the Brazilian region, we analyze ionospheric spatial decorrelation using ionospheric delay estimates generated by the Long-Term Ionospheric Anomaly Monitor (LTIAM) software package [3]. The processing summary is as follows. GPS data from subsets of the Brazilian reference stations with short separations between them (e.g., 100 km) is chosen and processed to precisely compute ionospheric delays and gradients. The tool then automatically investigates any ionospheric gradient candidates exceeding a threshold. The threshold is set large enough to be potentially hazardous to GBAS users (e.g., 200 mm/km). The selected ionospheric gradient candidates are passed through automated “false-alarm” screening and are then manually validated to discriminate real anomalies from faulty candidates.

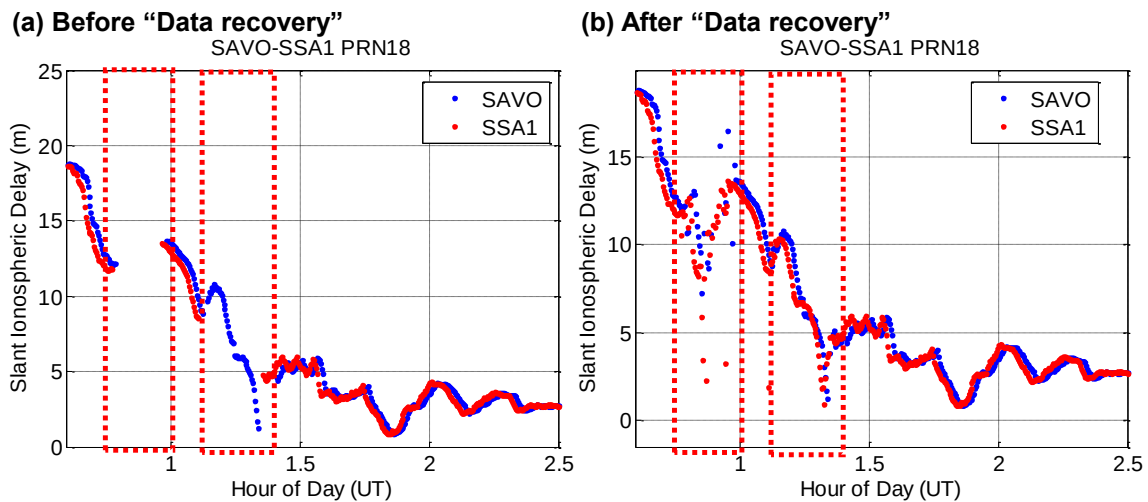


Figure 2. Dual-frequency estimates of slant ionospheric delays observed from stations SAVO (blue dot) and SSA1 (red dot) viewing PRN 18 on 14 November 2013 before "data recovery" (2a, left) and after "data recovery" (2b, right). Data loss from 00:47:30 UT to 01:26:30 UT is denoted by the red dashed boxes.

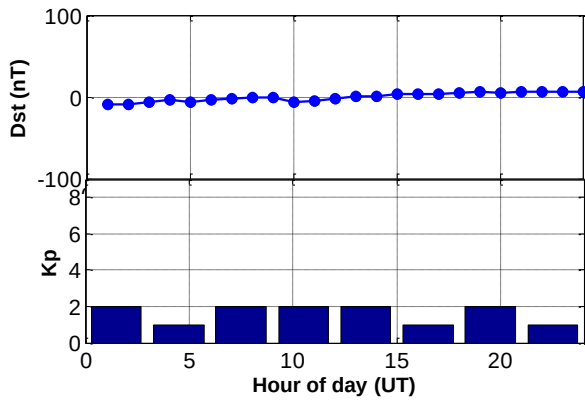


Figure 3. Variations of Kp and Dst indices on 31 December 2013.

The LTIAM detects obvious cycle slips of raw data and levels carrier-phase measurements using code measurements through pre-processing [3]. In the Brazilian region, however, intense ionospheric scintillations cause a large number of cycle slips in carrier-phase measurements. This causes the simple truth process to remove a significant number of measurements through short-arc and outlier removals, thus potential ionospheric gradients may not be detected. This is the motivation behind adding the new "data recovery" process in the manual validation step of the LTIAM. This "data recovery" process skips short-arc and outlier removals if there appears to be a large ionospheric spatial gradient in the data that would have been removed.

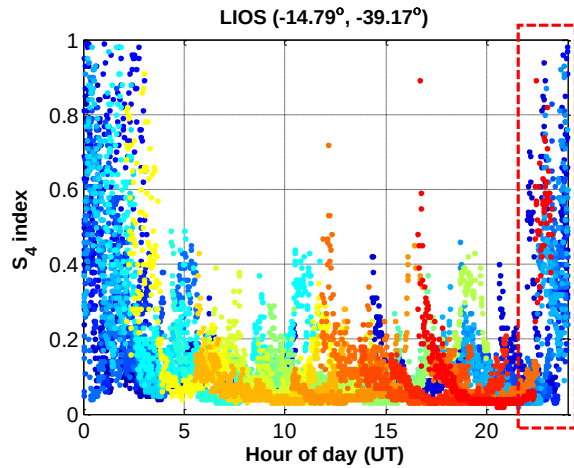


Figure 4. Time series of S₄ indices of all GPS satellites observed from LIOS (14.79°S, 39.17°W). The different colors of dots indicate S₄ data from different GPS satellites.

Figure 2 shows dual-frequency carrier-phase estimates of slant ionospheric delays observed from reference stations SAVO (blue dot) and SSA1 (red dot) viewing PRN 18 on 14 November 2013 before "data recovery" (2a, left) and after "data recovery" (2b, right). Data loss is observed in the carrier-derived estimate from 00:47:30 UT to 01:26:30 UT (in the two red dashed boxes) during vigorous ionospheric scintillation, as shown in Figure 2a. This is possibly caused by EPB events on this day. Since

many extreme ionospheric gradients are likely to be observed under the influence of EPBs, lost ionospheric data needs to be recovered to the extent possible in order to correctly assess the Brazilian ionospheric behavior. In Figure 2b, most slant ionospheric delays in the red boxes are recovered adequately, but some outliers appear. These outliers can be ignored through personal judgment because the "data recovery" process is in the manual step of LTIAM processing.

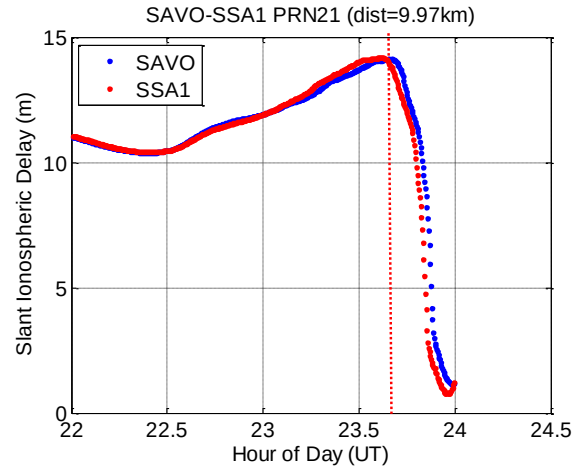


Figure 5. Dual-frequency estimates of slant ionospheric delays observed from stations SAVO (blue dot) and SSA1 (red dot) viewing PRN 21 on 31 December 2013. The red dashed line in vertical indicates the approximate start time (23:38:30 UT) of EPB-induced ionospheric delay drop.

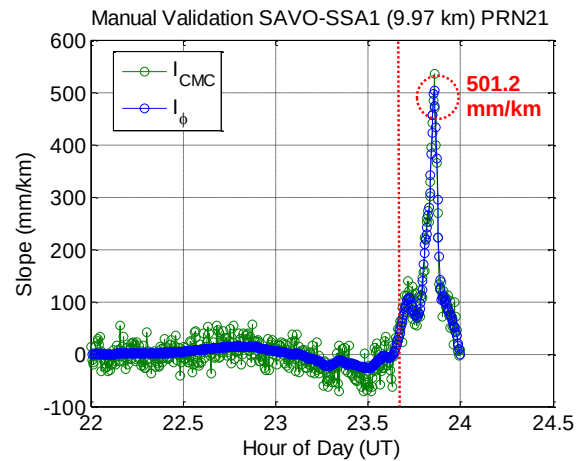


Figure 6. Dual-frequency estimates of ionospheric spatial gradient (blue circle) and L1 CMC estimates of ionospheric spatial gradient (green circle) observed from stations SAVO and SSA1 viewing PRN 21 on 31 December 2013. The maximum value in the red dashed circle is estimated as 501.2 mm/km at 23:38:30 UT. The red dashed line in vertical indicates the approximate start time (23:38:30 UT) of EPB-induced ionospheric spatial gradient.

We identified 229 extreme ionospheric spatial gradient candidates using "data recovery"-augmented LTIAM processing, and also verified them via "manual validation

with single frequency measurements.” Among those, four candidates are larger than 500 mm/km, implying that the upper bound on spatial gradients for the Brazilian model may need to be elevated to a considerably higher value compared to that of 425 mm/km in the CONUS (mid-latitude) model. Since GBAS simulates the worst-case ionospheric user errors in the position domain and mitigates them by inflating broadcast integrity parameters based on the threat model, a higher bound would induce an increase in the level of inflation, resulting in a reduction of system availability and potentially other problems. Thus, the largest gradient candidates must be validated thoroughly through the second-phase validation procedure in order to discriminate real gradients from faulty identifications caused by receiver glitches, satellite faults, or post-processing errors.

This paper selects the extreme gradient of 501.2 mm/km on 31 December 2013 from among the candidates as an example to demonstrate both the first-phase and second-phase validation procedures. On 31 December 2013, the geomagnetic condition was quiet since the highest geomagnetic indices, K_p and Dst , are 2 (low) and -9 nT (quiet), respectively, as shown in Figure 3, while high ionospheric scintillation activity was observed in the eastern part of the Brazilian region. Figure 4 shows the time series of S_4 indices for all GPS satellites observed from LIOS (14.79° S, 39.17° W). The different colors of dots indicate S_4 data collected from different GPS satellites. The S_4 index is defined as the standard deviation of the received signal power normalized to the average signal power at L-band and is an index of the strength of ionospheric scintillation. During the daytime (about 08:00:00 – 21:00:00 UT), S_4 indices varied between 0.0 and 0.5 according to satellite elevation angle (typically, higher values of S_4 occur at lower elevation angles due to weaker signal strengths). After sunset (about 21:00:00 UT), S_4 indices increase above 1.0, indicating that highly intense ionospheric scintillations were present and were possibly caused by post-sunset EPBs occurring in the eastern part of the Brazilian region.

Figure 5 shows dual frequency estimates for slant ionospheric delays observed from two stations, SAVO (12.93° S, 38.43° W) and SSA1 (12.97° S, 38.51° W), viewing PRN 21 on 31 December 2013. The delays at SAVO and SSA1 are denoted by red dots and blue dots. These two stations are also located in the eastern part of the Brazilian region, meaning that they also may be under the influence of EPBs. SSA1 first experiences ionospheric delay drop at about 23:38:30 UT (denoted by vertical red dashed line). SAVO, which lies 9.97 km northeast of SSA1, is affected shortly afterward. The ionospheric spatial gradient is estimated by dividing the difference in the apparent delay by their separation distance, as shown in Figure 6. It is noteworthy that the ionospheric spatial gradient (blue circle) increases up to 501.2 mm/km (within the red dashed circle) at 23:51:30 UT at a satellite elevation angle of 38.25 degrees as the delay at SSA1 is

depleted to 2.8 m, while the delay at SAVO remains at 7.28 m (but still within the depletion zone).

The dual-frequency carrier-phase estimates (blue circle) are compared with the single-frequency estimates (green circle) derived from L1 Code-Minus-Carrier (CMC) divergence as shown in Figure 6. These L1-only measurements are not as easily corrupted by cycle slips and data outages. The single-frequency ionospheric estimates exhibit a similar pattern to that of the dual-frequency estimate while there are cycle slips in both data sets (after 22:00 UT). Based on this agreement of the two estimates over time, it is reasonable to expect that this extreme ionospheric gradient candidate is real. However, we should further investigate the possibility of a faulty ionospheric candidate due to receiver-instigated events, satellite fault, or post-processing errors.

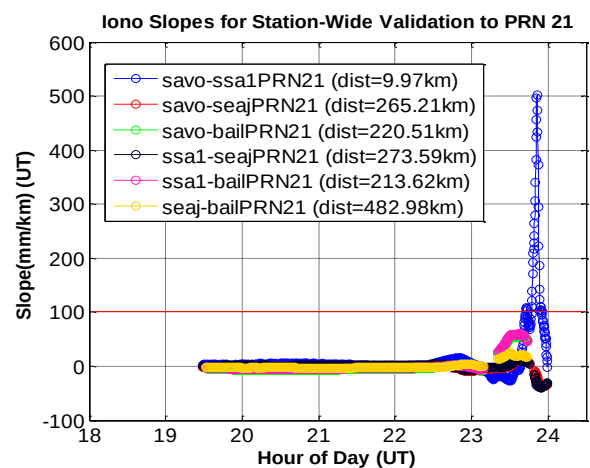


Figure 7. Dual-frequency estimates of ionospheric spatial gradients from all possible station pairs of SAVO, SSA1, SEAJ, and BAIL viewing PRN 21.

3.2 SECOND-PHASE VALIDATION PROCEDURE: LIMITATIONS IN THE BRAZILIAN REGION

In this paper, “station-wide validation” compares simultaneous observation data of ionospheric spatial gradients from nearby stations viewing the same GPS satellite. If the ionospheric spatial gradients from other stations in close proximity exhibit similar patterns and severities to the station pair observing the extreme ionospheric gradient candidate, the observed event is more likely caused by the ionosphere and not by receiver-instigated events. Continuing the analysis of the event shown in Figures 5 and 6, we selected the stations SEAJ and BAIL with their respective separation distances of about 270 km and 215 km from the mid-point of the SAVO/SSA1 pair. The ionospheric gradients observed from all possible station pairs viewing PRN 21 are represented in Figure 7. Except for the SAVO/SSA1 pair with a separation distance of 9.97 km (blue circle), the gradients of the other station pairs whose separation distances are a few hundred kilometers remain in a range of about 0 - 100 mm/km (100 mm/km is denoted by the red horizontal line). This suggests that separation

distances among the sparsely distributed Brazilian reference stations are too distant compared to the spatial scale of EPBs, which typically have a width of a few to several tens of kilometers in the east-west direction.

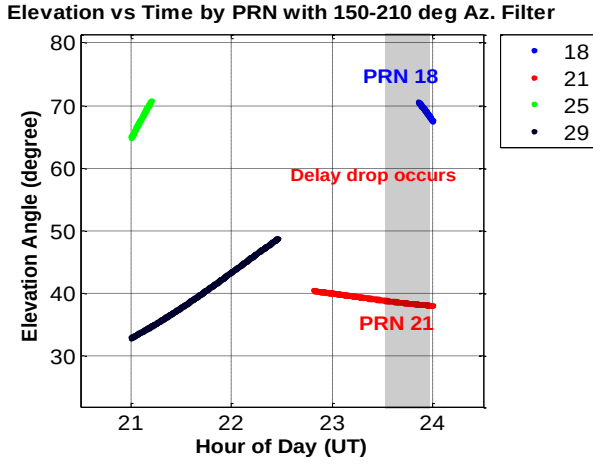


Figure 8. Elevation in degrees as a function of UT hour for satellites whose azimuth angle is within 150 to 210 degrees at SAVO and SSA1. Each color indicates the elevation angle of a different GPS satellite. The delay drop on PRN 21 occurs in the gray shaded area

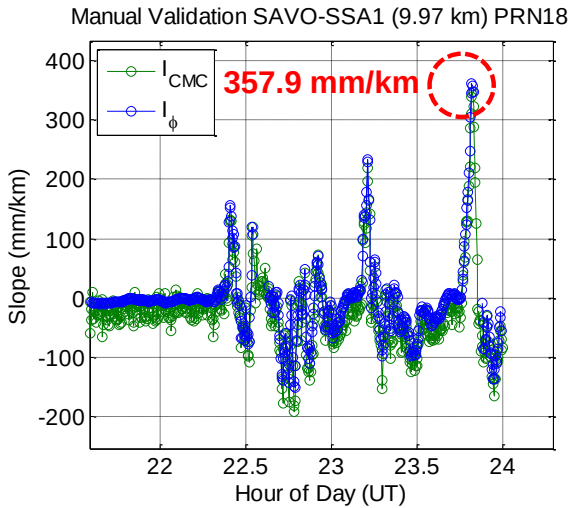


Figure 9. Dual-frequency estimates of ionospheric spatial gradient (blue circle) and single frequency estimates of ionospheric spatial gradient (green circle) observed from stations SAVO and SSA1 viewing PRN 18 on 31 December 2013. The maximum value in the red dashed circle is estimated to be 357.9 mm/km at 23:51:60 UT.

Another satellite passing across the same ionospheric region gave us further observations to help validate this extreme ionospheric gradient candidate. When the ionospheric delay drop occurs, the azimuth angle of PRN 21 is about 180 degrees. Accordingly, we searched for visible satellites observed from SAVO and SSA1 (as a function of elevation angle) whose azimuth angle is between 150 and 210 degrees using the azimuth filter [1], as shown in Figure 8. Elevation angles of one satellite tracked by two stations are given as the same color.

During the time of the delay drop on PRN 21, denoted by the gray shaded area, PRN 18 travels through a similar azimuthal region, but its elevation angle is considerably higher than that of PRN 21. Since the EPB is known to be stretched nearly from south to north with a scale of up to ~ 1000 km, the satellites observed from one station whose azimuth angles are similar to each other, are likely to be affected by the same EPB regardless of their elevation angles. The two stations SAVO and SSA1 viewing PRN 18 also observe a very large gradient of 357.9 mm/km at 23:51:00 UT, as shown in Figure 9. This suggests that it is more likely to be affected by the same EPB event as PRN 21, rather than by a single satellite fault or a different EPB.

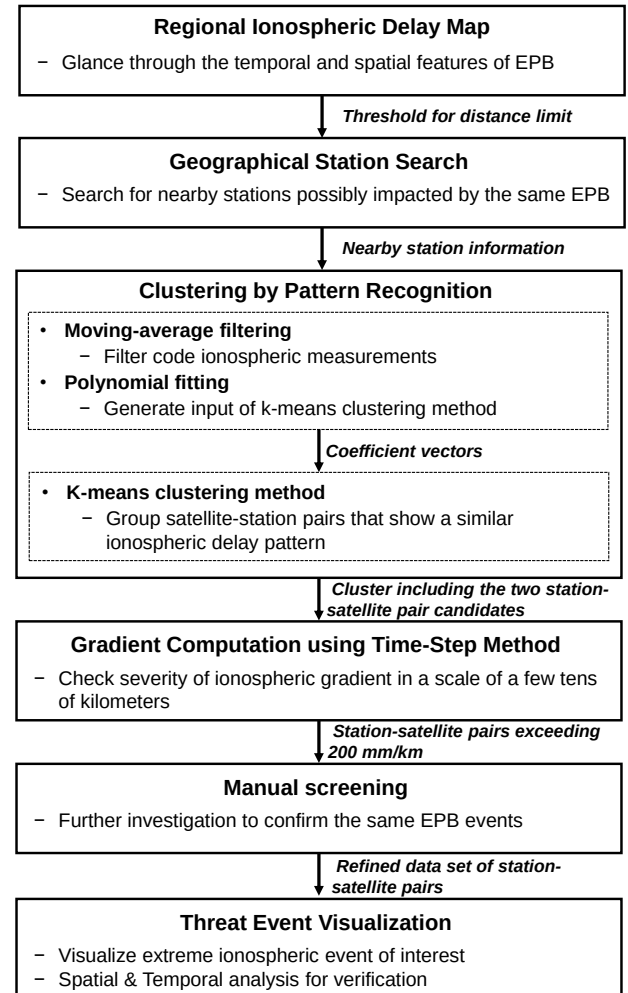


Figure 10. Threat Event Visualization Methodology

In the second phase of the validation procedure, we further investigated the possibility of faulty candidate caused by reference station faults or single satellite faults. Since anomalous ionospheric gradient was also observed from PRN 18 which passes through a similar ionospheric region to PRN 21, it is plausible that the observed gradient candidate here is not due to a single satellite fault, satellite bias removal error, or combinations of these causes. However, the “station-wide check” has not excluded receiver-instigated events due to a smaller scale of EPB or the sparse reference station distribution. This is

further motivation for us to develop and employ a new validation methodology for verifying an ionospheric gradient associated with EPBs in the Brazilian region.

4.0. NEW VALIDATION METHODOLOGY BASED ON TIME-STEP METHOD

The new ionospheric validation methodology, which utilizes multiple satellites and stations, is designed to find delays that show a similar pattern in order to check the severity of the observed gradient and to spatially and temporally visualize EPB events using all available data sources. For these purposes, the procedure consists of six steps, as shown in Figure 10: regional ionospheric delay map analysis, geographical station search, clustering of stations by pattern recognition, gradient computation using the time-step method, manual screening, and threat event visualization. Dual-frequency code and carrier-phase ionospheric estimates from the station pair observing an extreme ionospheric spatial gradient to be validated are taken as inputs to the verification procedure.

Analysis using a regional ionospheric delay map searches through the stations affected by unusual ionospheric behavior and locates the EPB event. Based on this information, nearby GPS reference stations are sought and are returned to the clustering step. The “K-means clustering method” defined in [5] groups station-satellite pairs that exhibit similar delay patterns. The cluster including two station-satellite pairs which observe extreme gradient to be validated is used for the next step. The ionospheric spatial gradients of the station-satellite pairs in this cluster are computed using the “time-step method.” Station-satellite pairs from this subset whose gradient exceeds a threshold are passed to the manual screening step for further investigation. Once the first five steps identify an apparently anomalous set of data, we verify the ionospheric gradient candidate using a visualized EPB event in a different time and location in conjunction with the location of stations and satellite Ionospheric Pierce Point (IPP) tracks at which the line of sight between satellite and receiver and the thin-shell ionosphere intersect. The details of each step are described in the following subsections.

4.1 REGIONAL IONOSPHERIC DELAY MAP ANALYSIS

It is known that most EPBs occupy a relatively small region compared to other large-scale ionospheric anomalies such as Storm Enhanced Density (SED), Equatorial Ionospheric Anomaly (EIA), and so on. Since the EPB has a fan-like structure stretched from south to north, the zonal width (east-west direction) is relatively small with a scale of a few tens of kilometers [2]. Therefore, knowledge of the geographical range and temporal coverage of an EPB event, such as location, time, width, level of depletion, and moving distance, is important so as to avoid unnecessary effort in analyzing station-satellite pairs that are not affected by the same EPB event.

The region of interest in this methodology is in the EPB depletion region where ionospheric irregularities are the most vigorous. Cycle slips during intense ionospheric scintillation are likely to corrupt carrier-phase measurements but not code measurements. Thus, we utilize code measurements to build the regional ionospheric map. Highly noisy code measurements can degrade ionospheric delay map accuracy, so a “moving average” filter is used to suppress the noise level. The moving average filter produces each value in the output signal (moving-average-filtered code estimates of slant ionospheric delays) by averaging a number of points from the input signal (code estimates of slant ionospheric delays) within a given time window [6]. The time window of 7.5 minutes is chosen for the filter in this paper. This is to meet the competing requirements of reducing noise in code measurements and capturing a rapidly varying ionospheric trend.

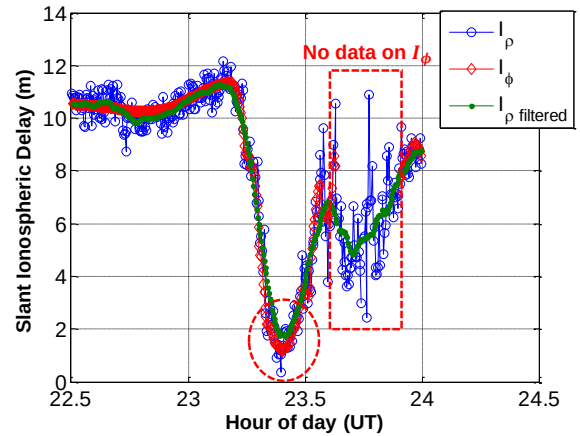


Figure 11. Dual-frequency code estimates of slant ionospheric delay (I_ρ ; blue-dashed circle), dual-frequency carrier-phase estimates of slant ionospheric delay (I_ϕ ; red-dashed diamond), and 7.5 min moving-average-filtered code estimates of slant ionospheric delays ($I_{\rho \text{ filtered}}$; green-dashed dot) of PRN 18 observed from BATF.

Figure 11 shows the time series of dual-frequency code estimates of slant ionospheric delay (I_ρ ; blue dashed circle), dual-frequency carrier-phase estimates of slant ionospheric delay (I_ϕ ; red dashed diamond), and 7.5 min moving-average-filtered code estimates of slant ionospheric delay ($I_{\rho \text{ filtered}}$; green dashed dot) of PRN 18 observed from the station BATF (17.55° S, 39.74° W) on 31 December 2013. This station experiences a steep slant ionospheric delay drop of about 11 m due to EPB-induced depletion from about 23:17:00 UT. During this depletion, the long-term trend of code-derived estimates is well captured by the moving-average-filtered code estimates of slant ionospheric delay. After the first delay drop, data outages appear between 23:37:30 UT and

23:54:00 UT in the carrier-derived estimate (inside the red box). This is caused by frequent cycle slips during EPB-induced scintillation. However, only a few data points in the moving average filtered code delay are corrupted temporarily during intense ionospheric scintillation. This improves the observability of the EPB in the regional delay map. On the other hand, moving-average-filtered code delay contains a smoothing effect during rapid delay depletion, resulting in an underestimation of delay (nearly 0.5 m), as shown in the red dashed cycle. This error is tolerable since this quantity is only used for the validation of a threat.

After computing the moving-average-filtered code estimates of slant ionospheric delays of all GPS satellites from all available GPS reference stations, these are converted into equivalent moving-average-filtered code estimates of vertical delays using a geometric mapping function for which an ionosphere thin shell model with a hypothetical height of 350 km is assumed [7]. The geometric mapping function M used in this paper is defined as below [8].

$$M(el, h_t) = \left\{ \cos \left[\sin^{-1} \left(\frac{R_e \cos(el)}{R_e + h_t} \right) \right] \right\}^{-1} \quad (1)$$

where R_e is the radius of Earth (6378 km), h_t is the height of the ionospheric shell (350 km), and el is the elevation angle of the line of sight between a receiver and satellite. The color contour map with a temporal resolution of 30 sec is created by interpolating these scattered IPPs with their vertical delays over a 1-by-1 degree rectangular grid using a triangle-based cubic interpolation method [9].

Using the regional ionospheric map, we glance through the ionospheric behavior and locate the EPB event. Based on information on the occurrence time and location of the EPB, the occurrence time of observed extreme gradient to be validated, and the locations of the two stations, we determine a criterion of distance for “geographical station search” in the next step.

4.2 GEOGRAPHICAL STATION SEARCH

The validation algorithm automatically searches for nearby stations possibly impacted by the same EPB that generated a severe gradient candidate requiring validation. For a criterion of distance from a station pair of the gradient candidate, this step utilizes a threshold predetermined from the regional delay map analysis. Stations in close proximity to the station pairs observing the gradient candidate would experience similar ionospheric behavior, such as steep delay drops and extreme spatial gradients, in the presence of an EPB event.

4.3 CLUSTERING BY PATTERN RECOGNITION

If an actual EPB hits lines of sight from multiple stations to satellites moving in a similar direction, the ionospheric delays of all station-satellite pairs will experience approximately the same amount of delay drops. Accordingly, it is reasonable to confirm an actual ionospheric event by investigating similar delay patterns and showing severe ionospheric gradients among given station-satellite pairs during the ionospheric anomaly validation process. However, grouping station-satellite pairs affected by the same EPB event via manual inspection is a time-consuming task. In this step, we thus implement a well-known pattern recognition technique known as “k-means clustering” explained in [5] to automate the selection of station-satellite pairs from the chosen nearby stations showing a similar delay pattern.

This process was already used in automated velocity estimation of ionospheric front that is assumed to be spatially linear and be semi-infinite for CONUS [10]. Since the EPBs has approximately linear wave front in south-north direction, we use this process to group station-satellite pairs that exhibit similar delay patterns. Previous work used 31 real coefficients obtained from a 30th order -polynomial curve fitted to a time series of dual-frequency estimates of slant ionospheric delays as the feature vectors which represent some objects [5][10]. However, these were intended for mid-latitude ionosphere, which is less varying and fluctuating than that of the equatorial region. In particular, various scales of ionospheric irregularities within the EPB induce many different types of ionospheric delay curve features.

Thus, in this paper, we determined the optimal order for polynomial fitting as 40th (41 real coefficients) to minimize Root-Mean-Square-Errors (RMSEs) between actual ionospheric delay and polynomial-fitted delay. The higher order (as compared to that in mid-latitudes) is attributed to EPB-induced rapidly varying ionospheric curves. In addition to the order of the polynomial fit, we use the moving-average-filtered slant ionospheric delays for polynomial fitting (as explained in Subsection 4.1), because there is a considerably higher possibility that cycle slips and data outages occur in carrier-derived estimates during EPB-induced scintillation.

A k of 5, which is the number of clusters into which data is divided, is chosen. For applying the “k-means clustering” method, two station-satellite pairs, which observe an extreme ionospheric gradient to be validated are taken as inputs. The automated clustering process assigns ionospheric delays of all possible GPS satellites tracked by the reference stations screened by “geographical station search” (see Subsection 3.2) to five clusters. This number of clusters is larger than that in [10] because complex patterns of ionospheric delays caused by various scales of ionospheric irregularities within the EPB should be grouped more delicately by using more clusters. In spite of this benefit, great care must be taken in the use

of more clusters because it leads to loss of ionospheric data that could otherwise be used for validation. Among five clusters, a cluster including two station-satellite pairs that observe the extreme ionospheric gradient to be validated is used in the following procedures.

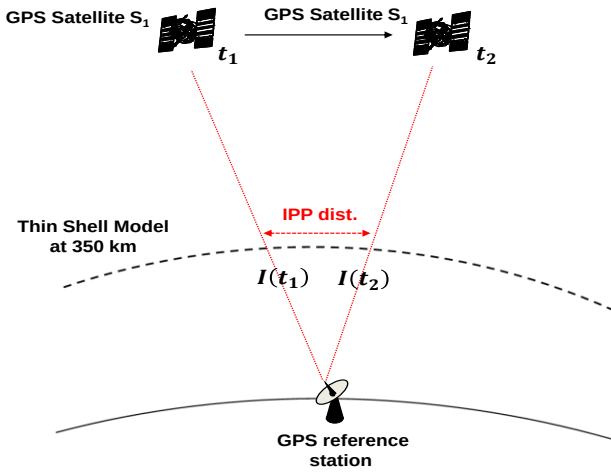


Figure 12. Satellite and receiver configuration for the time step method [11].

4.4 GRADIENT COMPUTATION USING TIME-STEP METHOD

In this step, ionospheric gradients of all station-satellite pairs from the selected cluster are computed using the “time-step method.” This method was originally developed in [4] to obtain sufficient ionospheric spatial gradients at distances less than the physical separations of reference stations. One GPS satellite and one reference station are formed as a pair for gradient computation, as shown in Figure 12. A difference of ionospheric delays ($I(t_1)$ and $I(t_2)$) between one epoch (t_1) and a later epoch (t_2) is divided by the travel distance of the IPP from t_1 and t_2 to compute the gradient [11]. As we are differencing two delays of the same station-satellite pair, both the remaining satellite and receiver IFB errors are nearly removed. We choose a time window of one minute ($\Delta t = t_2 - t_1$). This time window is useful to investigate localized ionospheric spatial decorrelation in a scale of up to a few tens of kilometers when the IPP moves with a velocity of up to 300 m/s at low elevation (e.g., 10 degrees), assuming a hypothetical 2-D ionospheric shell height of 350 km. Note that estimated gradients using this method also contain temporal (as well as spatial) decorrelation errors. However, this error is tolerable since the gradients obtained using this method are only used for the validation of the ionospheric anomaly event.

Ionospheric gradient computation using the time-step method is performed on carrier-derived estimates, which are more accurate than code-derived estimates. However, carrier-derived estimates may contain an excessive number of cycle slips, especially when the intensity of ionospheric irregularities is high. This may lead to a

faulty ionospheric spatial gradient which is large due to the short IPP distance. Thus, the automated process computes the gradient using the time-step method only when no cycle slip occurs in the carrier-derived estimates. Station-satellite pairs whose ionospheric gradients exceed a threshold (currently 200 mm/km in the slant domain) are further investigated to confirm that they are affected by the same localized ionospheric anomaly.

4.5 MANUAL SCREENING

It is not uncommon that there are multiple EPBs with zonal separation distances of a few hundred kilometers in a single GPS field of view [2]. Accordingly, the automated pattern recognition process may not be perfectly able to discriminate station-satellite pairs affected by the same EPB from other station-satellite pairs affected by another EPB. Manual screening is used to re-examine sets of data from the “gradient computation using time-step method” (see Subsection 4.4) to determine whether the resulting moving-average-filtered code-estimates of slant ionospheric delays appear to be produced by the same EPB event. However, some degree of personal intervention is involved due to various unique cases. If the ionospheric delay pattern of a particular station-satellite pair is akin to those of two station-satellite pairs observing an ionospheric gradient to be validated, this station-satellite pair is used for threat event visualization.

4.6 THREAT EVENT VISUALIZATION

Once the automated procedure acquires an apparently anomalous set of data based on the similarity of ionospheric delay patterns and the severity of ionospheric gradients, a visualization of the EPB event is performed using the refined data set. We visualize the observed ionospheric gradient event in a time series in conjunction with the satellite IPP tracks, the presumed motion of the EPB, and the location of the affected reference stations. Two-dimensional regional ionospheric delay maps in gray scale are utilized in order to track the movement of the EPB front visually. Next, we plot the gradients of station-satellite pairs in the refined data set, computed using the time-step method, at the corresponding locations of IPPs over the regional ionospheric delay map. These two sub-steps are repeated every 30 sec sequentially over the time period when the EPB affects the selected station-satellite pairs.

This visualization enables us to scrutinize the relationship between the EPB event and the observed extreme ionospheric spatial gradient in both the time domain and the spatial domain, taking into account the EPB and IPP motion, station locations, and the occurrence times of large gradients. When this comprehensive analysis supports the fact that multiple station-satellite pairs were impacted by the same EPB at different times and locations, the extreme ionospheric gradient candidate of interest is finally validated as real

Table 2. Summary of threat event visualization

	Number of station-satellite pairs
Total number in the Brazilian region on 10 December 2013	2871
Geographical station search (< 1500 km)	1706
Clustering by pattern recognition ($k=5$)	152
Gradient computation using time-Step method (> 200 mm/km)	78
Manual Screening	6

5.0. THREAT EVENT VISUALIZATION RESULTS

In Section 3.0, the observed ionospheric gradient candidate was verified through the two-phase procedure. Re-examination of dual-frequency carrier-phase estimates using “manual validation with single frequency measurements” confirmed no possible corruptions due to post-processing errors and receiver faults during the intense scintillation. The “satellite-wide check” showed that the observed gradient candidate was not caused by a single satellite fault or bias error. However, a slight doubt remains that the actual EPB induces the extreme ionospheric gradient since we could not exclude any receiver-instigated event via the “station-wide check.” In this section, we obtain further support for the conclusion that the observations of an extreme ionospheric gradient of 501.2 mm/km are due to an actual ionospheric event. We do this via the newly developed methodology based on the “time-step method” that uses multiple widely distributed stations and satellite pairs. The results from the analysis using the above methodology in Section 4.0 are summarized in Table 2, and the details are as follows.

Figure 13 shows two-dimensional regional ionospheric delay maps on 31 December 2013. The color scales of vertical ionospheric delay are between 0 and 10 m. The region of interest, where the EPB affects the theoretical

350-km-altitude IPPs (black rectangles) of PRN 21 observed from SAVO and SSA1, is denoted by the red dashed box. The EPB with a delay depletion of about 7 m occurs at about a latitude of 17.8° S and a longitude of 42.0° W as identified by the black arrow. The IPPs are about 310 km east of the EPB at 23:00 UT, staying in the middle of the enhanced delay region as shown in Figure 13a. This corresponds closely with the result of Figures 5-6 in that no delay drop and no extreme ionospheric gradient are observed at the same time.

The locations of the station pairs and the occurrence location of the EPB are used so as to compute a threshold or distance limit of “geographical station search.” The horizontal separation distance between the location of the mid-point of SAVO and SSA1 and the occurrence location of the EPB is measured as about 800 km. Furthermore, each GPS station observes the ionospheric thin shell at an altitude of 350 km within a radius of about 600 km when using a satellite at elevation angle above 30 degrees (lower satellite elevation angle induces low quality of ionospheric measurement). Based on the separation distance and the ionospheric observability of the reference station, we conservatively set the threshold as 1500 km, considering the case that the EPB is observed by the reference stations located in the western part of the EPB occurrence location.

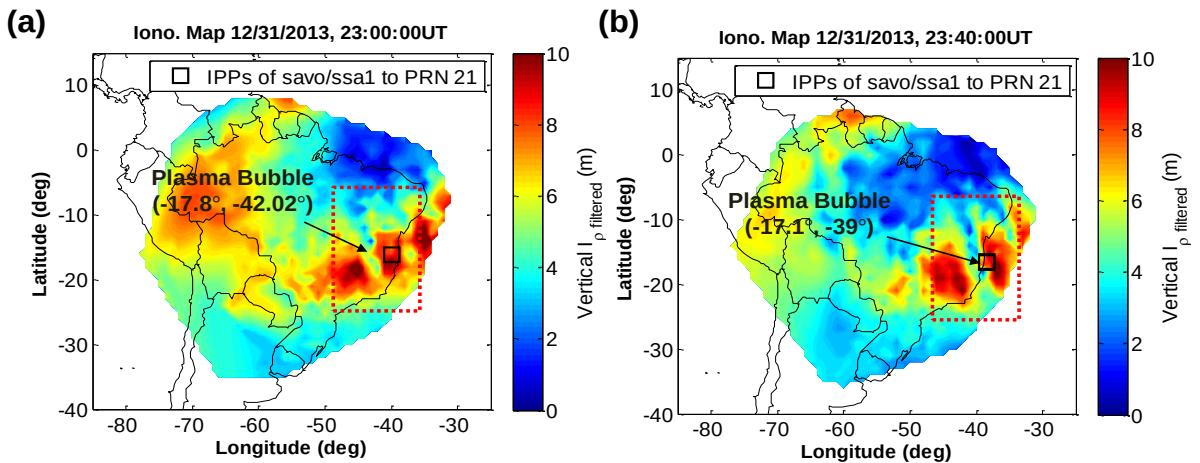


Figure 13. Two-dimensional regional ionospheric delay map at 23:00:00UT and at 23:40:00 UT on 31 December 2013. The color scales of vertical ionospheric delay are between 0 and 10 m. The EPB affecting IPPs (black rectangles) of PRN 21 observed from SAVO and SSA1 is denoted by the red dashed box.

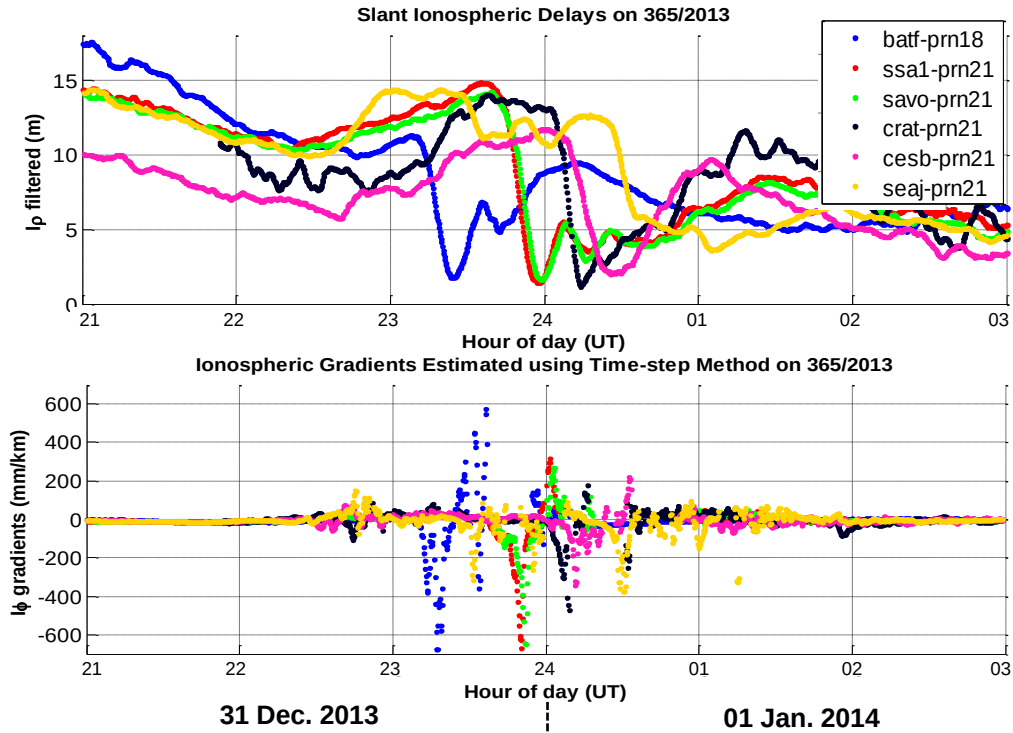


Figure 14. Moving-average-filtered code estimate of slant ionospheric delay (top) and dual-frequency estimates of ionospheric spatial gradient computed using time-step method (bottom) for BATF-PRN 18 (blue), SSA1-PRN 21 (red), SAVO-PRN 21 (green), CRAT-PRN 21 (black), CESB-PRN 21 (pink), SEAJ-PRN 21 (yellow) as a function of time.

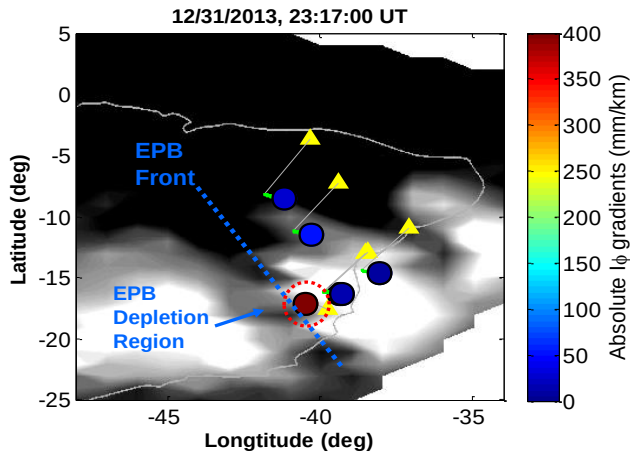
In Figure 13b, it is obviously seen that the EPB moves further eastward and affects the IPPs, maintaining a fan-like structure stretched from southeast to northwest (due to magnetic declination) at 23:40:00 UT. At this time, each station undergoes steep delay drops of 14 m. The eastward velocity of the EPB is estimated as about 147 m/s by comparing depletion areas at different times. In addition, IPPs move with a velocity of 71 m/s at an azimuth angle of 89 degrees, namely nearly eastward movement. Based on this information, the high rate of downward change of ionospheric delay within the EPB comes up with the IPPs with a higher eastward velocity for about 40 min after the occurrence, and this may lead to the extreme ionospheric spatial gradient. This provides further support to conclude the observed extreme ionospheric spatial gradient candidate caused by the localized ionospheric anomaly.

Table 3. The station-satellite pairs from west to east and separation distances from the midpoint of two station SAVO and SSA1

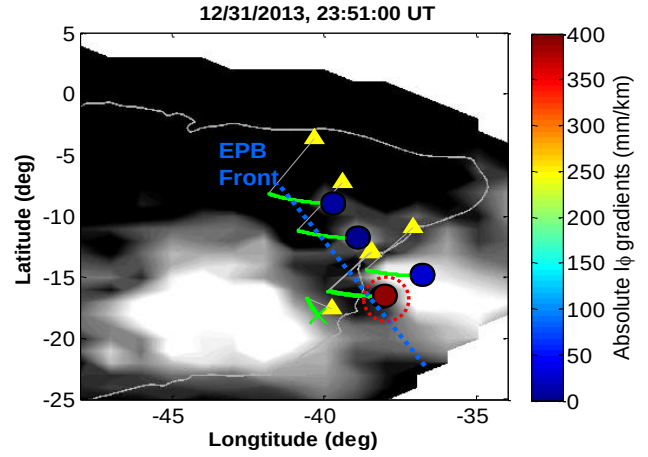
Station-satellite pair from east to west	Separation distance
CESB (3.68° S, 40.33° W) - PRN 21	1045 km
CRAT (7.24° S, 39.42° W) - PRN 21	640 km
BATF (17.55° S, 39.74° W) - PRN 18	526 km
SEAJ (10.92° S, -37.10° W) - PRN 21	269 km

The algorithms took the SAVO-PRN21 and SSA1-PRN21 pairs as an input and searched for nearby stations within a radius of 1500 km centered at the midpoint of the station pair. The returned number of stations was 73. In the “clustering by pattern recognition,” the moving-average-filtered code estimate of the selected 1706 station-satellite pairs during the nighttime were processed, in order to choose the cluster that exhibits a similar delay pattern to those of SAVO-PRN21 and SSA1-PRN 21. Next, the ionospheric gradients of the resulting 152 station-satellite pairs based on the similarity of ionospheric delay pattern, were computed using the time-step method, and 78 candidates exceeded a threshold of 200 mm/km. Note that the automated screening based on the pattern recognition is not faultless – some station-satellite pair candidates that exhibit slightly different delay pattern pass through it. Thus, we performed manual screening on the remaining candidates, and six candidates (including SAVO-PRN 21 and SSA1-PRN 21) were finally confirmed to compose the refined data set for threat event visualization as summarized in Table 3.

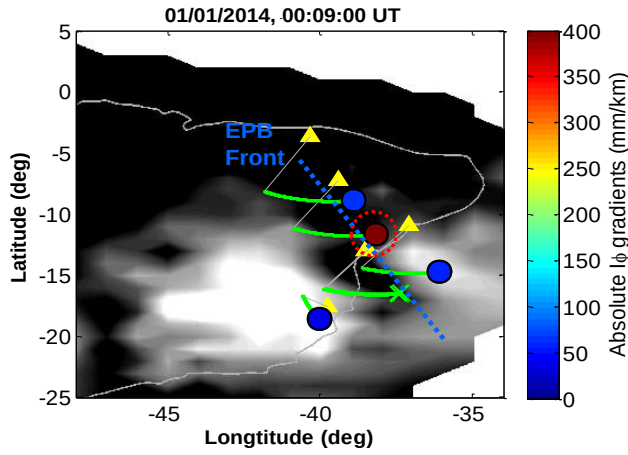
(a) BATF-PRN 18



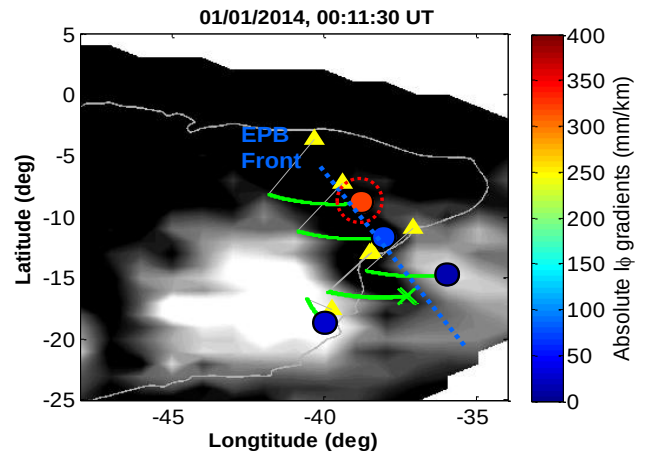
(b) SSA1/SAVO-PRN 21



(c) CRAT-PRN 21



(d) CESB-PRN 21



(e) SEAJ-PRN 21

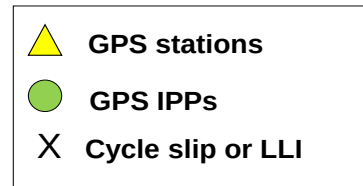
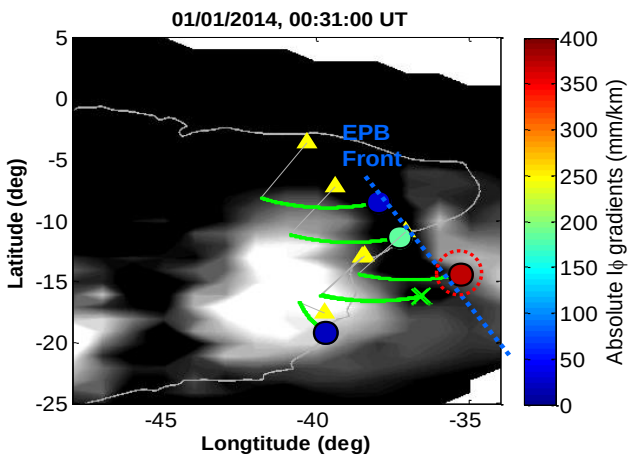


Figure 15. Threat event visualization of 10 December 2013 EPB event in the eastern part of the Brazilian region from 23:00:00 UT on 31 December 2013 to 01:00:00 UT on 01 January 2014. Figure 15a-e show the EPB, the locations of the station, the IPPs with their gradients at the time when the maximum gradient of each station-satellite pair is observed. The EPB-induced depletion region is pointed by the blue arrow in Figure 15a, and this front is drawn with blue dashed line. The values of ionospheric gradients are mapped at its IPP locations (circle) connected with the locations of stations (yellow triangle) except for cycle slips (cross). The IPP tracks in the previous time are drawn with green solid line. The IPP impacted by the EPB is denoted by the red dashed circle. Note that the color scales of absolute ionospheric gradients are between 0 and 400 mm/km.

Figure 14 shows the moving-average-filtered code estimates of slant ionospheric delay (top) and dual-frequency estimates of ionospheric spatial gradient computed using the time-step method (bottom) of the refined station-satellite pairs as a function of time. BATF-PRN 18 (blue), SSA1-PRN 21 (red), SAVO-PRN 21 (green), CRAT-PRN 21 (black), CESB-PRN 21 (pink), and SEAJ-PRN 21 (yellow) experience ionospheric delay drops of about 10-14 m in order from 23:17 UT, which exhibit similar delay patterns, resulting in all of the six extreme ionospheric gradients larger than 300 mm/km.

It is noteworthy that their order of occurrence time of delay drops and their order of occurrence time of the maximum extreme gradients appear not to be correlated with the longitudinal locations of reference stations. In other words, ionospheric depletion features do not occur at earlier times in lower longitudes (reference station located further west). This is possibly due to fact that the shape of an EPB is aligned north-south and slightly tilted east-west due to magnetic declination. For this reason, using only the information on occurrence time of large ionospheric gradients of the refined station-satellite pairs, we cannot conclude that these multiple station-satellite pairs are affected by the same EPB event.

The EPB-induced extreme ionospheric gradient event observed in the eastern part of the Brazilian region is visualized using all available data sources temporally and spatially from 23:00:00 UT on 31 December 2013 to 01:00:00 UT on 01 January 2014. Figures 15a-e show the EPB (indicated by the blue arrow), the locations of the stations (yellow triangles), and the IPPs (circles) with their gradients at the time when the maximum gradient of each station-satellite pair is observed. The black region indicates a lower value of vertical ionospheric delay while the white region indicates a higher value of vertical ionospheric delay. We assumed that the EPB front is linear as drawn with the blue dashed line. The values of ionospheric gradients computed using the time-step method are mapped at the corresponding locations of IPPs except when cycle slips (cross) occur. The IPP tracks in the previous time are drawn with a green solid line. The IPP impacted by the EPB front is denoted by the red dashed circle. Note that the color scales of absolute ionospheric gradients are between 0 and 400 mm/km.

The EPB front moving eastward first impacts the IPP of BATF-PRN 18 and this high rate of downward change of delay in the boundary layer of the EPB induces an extreme ionospheric gradient above 650 mm/km at 23:17:00 UT as shown in Figure 15a. The SSA1-PRN 21 and SAVO-PRN 21 experience EPB-induced delay drops of 14 m (see Figure 5) and extreme ionospheric gradients above 650 mm/km sequentially as the EPB front approaches them, while frequent cycle slips and data outages occur at BATF-PRN 18 due to the intense ionospheric irregularities within the EPB depletion region at 23:51:00 UT in Figure 15b.

Figures 15c and 15d show the time when the EPB front affects CRAT-PRN 21 (00:09:00 UT on 01 January 2014) and CESB-PRN 21 (00:11:30 UT on 01 January 2014), respectively. We can expect CESB which lies 406 km northwest of CRAT to be affected by the EPB moving eastward at an earlier time, but this does not actually happen. This is likely caused by the fact that the angle between the EPB front and the north vector (measured in a counterclockwise direction starting from the north vector) is larger than the angle between the line from CRAT to CESB and the north vector (measured in a counterclockwise direction starting from the north vector). The maximum gradients of both the station-satellite pairs are estimated as about 470 mm/km and about 340 mm/km, respectively. In Figure 15e, the EPB moves further eastward maintaining a fan-like structure stretched from southeast to northwest at 00:31:00 UT on 01 January 2014. Ionospheric delay is depleted by about 9 m at SEAJ-PRN 21, resulting in the maximum gradient of about 370 mm/km.

Based on a good consensus among the spatial and temporal characteristics of the EPB, the IPP tracks with their gradients, occurrence time of large gradients, and the location of stations, we verified that the satellite-station pairs spread over several hundreds of kilometers in the east-west direction were impacted actually by the same EPB moving eastward. Therefore, the extreme ionospheric gradient investigated in this paper is finally validated to be real.

6.0. CONCLUSION

In this paper, we have applied a two-phase procedure of verifying extreme ionospheric spatial gradients using a multi-dimensional approach in the Brazilian region. The procedure consisting of various validation methods discriminates real ionospheric spatial gradients multi-dimensionally from any false anomaly caused by receiver-instigated events, a single satellite fault, or post-processing errors. However, a relatively smaller scale of EPB and the sparse distribution of the Brazilian network stations make it difficult to use the existing validation methods as they are, and thus new techniques need to be employed. This paper proposes a validation methodology based on “time-step method.” This process utilizes multiple satellites and stations that exhibit a similar trend of ionospheric delay with that of the threat candidate. A time-step method enables us to estimate ionospheric gradients over any short baseline distance and thus compensates for the lack of ionospheric observability caused by a sparse distribution of the GPS reference stations. The methodology visualizes a time series of severe gradient events in conjunction with satellite IPP tracks, EPB motion, and ground station locations.

This paper verified an extreme ionospheric gradient above 500 mm/km during intense ionospheric scintillation using the multi-dimensional verification procedure. Coexisting ionospheric observation data from another satellite

passing across the similar region of the EPB event was anomalously as high as 360 mm/km. The comprehensive analysis using time-step validation demonstrated that multiple station-satellite pairs were impacted by the same EPB event at different times and location. Therefore, the extreme ionospheric gradient investigated in this paper has been validated to be real (due to an EPB). In addition, verification of other extreme events larger than 500 mm/km was also completed.

The proposed methodology will be utilized to develop an ionospheric anomaly threat model for GBAS operation in the Brazilian region. More work is also needed to develop proper mitigation strategies in the equatorial region that reduce the impact of extreme ionospheric spatial gradients on GBAS user availability in the Brazilian region. It is planned to conduct statistical analysis of the Brazilian ionospheric gradient data in various ways, such as time and regional distributions of ionospheric threat occurrence, vulnerability of station-satellite configurations to the EPB, and so on. The results from this study can be used when determining the worst-case ionospheric position error and to reduce unnecessary conservatism in the process of ionospheric threat mitigation.

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